2 - Coding Golf Assesment

Back

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Q1 **Bubbled Sorting** 25 marks

Write a C code for performing Bubbled Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
#include <stdio.h>

int main() {
    int A[5] = {8, 2, 3, 5, 7};
    int i, j, temp;

    // Bubble Sort
    for (i = 0; i < 5 - 1; i++) {
        for (j = 0; j < 5 - i - 1; j++) {
            if (A[j] > A[j + 1]) {
                temp = A[j];
                A[j] = A[j + 1];
                A[j + 1] = temp;
            }
        }
    }

    printf("Sorted array: ");
    for (i = 0; i < 5; i++) {
        printf("%d ", A[i]);
    }

    return 0;
}
```

Input

stdin input

Output

```
Sorted array: 2 3 5 7 8
```

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Back

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Q2 **Insertion Soring** 25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```c
#include <stdio.h>

int main() {
    int A[5] = {8, 2, 3, 5, 7};
    int i, j, key;

    for (i = 1; i < 5; i++) {
        key = A[i];
        j = i - 1;

        while (j >= 0 && A[j] > key) {
            A[j + 1] = A[j];
            j--;
        }

        A[j + 1] = key;
    }

    printf("Sorted array: ");
    for (i = 0; i < 5; i++) {
        printf("%d ", A[i]);
    }

    return 0;
}
```

Input

stdin input

Output

```
Sorted array: 2 3 5 7 8
```

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Back

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Q3 **Quick Sorting**

25 marks

Write a C code for performing Quick Sort - n=7 A[] = {8, 2, 3, 5, 7,1,6}.

Your Code

```c
#include <stdio.h>

void quickSort(int a[], int low, int high)
{
    int i, j, pivot, temp;

    if (low < high)
    {
        pivot = a[high];
        i = low - 1;

        for (j = low; j < high; j++)
        {
            if (a[j] < pivot)
            {
                i++;
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }

        temp = a[i + 1];
        a[i + 1] = a[high];
        a[high] = temp;

        int p = i + 1;

        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}

int main()
{
    int a[] = {8, 2, 3, 5, 7, 1, 6};
    int n = 7, i;

    quickSort(a, 0, n - 1);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Input

stdin input

Output

```
Sorted array: 1 2 3 5 6 7 8
```

2 - Coding Golf Assesment

Back

NS

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Q4 **Merger Sort**

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```c
#include <stdio.h>

void merge(int a[], int low, int mid, int high)
{
    int i = low, j = mid + 1, k = 0;
    int temp[100];

    while (i <= mid && j <= high)
    {
        if (a[i] < a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }

    while (i <= mid)
        temp[k++] = a[i++];

    while (j <= high)
        temp[k++] = a[j++];

    for (i = low, k = 0; i <= high; i++, k++)
        a[i] = temp[k];
}

void mergeSort(int a[], int low, int high)
{
    if (low < high)
    {
        int mid = (low + high) / 2;

        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);

        merge(a, low, mid, high);
    }
}

int main()
{
    int a[] = {8, 2, 3, 5, 7, 6, 4, 1};
    int n = 8, i;

    mergeSort(a, 0, n - 1);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Input

stdin input

Output

```
Sorted array: 1 2 3 4 5 6 7 8
```